Random Dimension Selection as a Strong Baseline: Training Paradigm Associates with the Value of Task-Aware Pruning

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Abstract

Modern text embedding models produce high-dimensional vectors (768–4096 dimensions) that are costly to store and compare at scale. Prior work has shown that randomly removing a large fraction of dimensions often causes only moderate performance loss, suggesting substantial redundancy. We revisit this observation in the stricter setting of post-hoc compression of frozen, off-the-shelf embedding models, and ask a narrower question: *when, and by how much, does task-aware dimension selection improve over random selection?* Across 11 embedding models and 35 MTEB tasks, random selection is a strong baseline, especially for contrastively trained models, where a task-aware oracle improves retention by only +2.2%–+5.0% at 256 dimensions (95–98% retention). However, the gap is substantially larger for non-contrastive models (BART-Base: +10.1%, Roberta-Large: +8.6%) and the task-specifically fine-tuned Roberta-InBedder (+8.2%), showing that interchangeability is **model-family dependent rather than universal**. Additional analyses reveal that the strength of this conclusion depends on compression ratio, importance definition, and evaluation protocol: cross-task transfer degrades at aggressive pruning (dim $\leq$ 64), Shapley-based importance reveals more structure than standalone scoring, and random selection is also more robust under domain shift. We conclude that random selection should be the default baseline for post-hoc embedding compression, but its adequacy depends on model family and compression regime.

1 Introduction

Dense text embeddings power search, clustering, and classification across modern NLP pipelines, but state-of-the-art models produce 768–4096 dimensional vectors whose storage and comparison costs scale linearly with dimensionality. A growing body of work has documented the redundancy of these embeddings: Takeshita et al. (2025) show that randomly removing 50% of dimensions causes only minor performance drops, while Tsukagoshi and Sasano (2025) find that keeping only 25% of dimensions often incurs negligible loss. These findings suggest **dimension importance is broadly distributed**—no small subset carries disproportionately more information.

We revisit this observation with a narrower question: *when, and by how much, does task-aware dimension selection improve over random selection?* This question has direct practical implications. If random selection is sufficient, then post-hoc embedding compression requires no task-specific evaluation, no gradient computation, and no retraining—making it trivially deployable. If task-aware selection provides substantial gains, then investing in importance estimation is worthwhile. The answer, we show, depends critically on *training paradigm*: contrastive training produces near-uniform importance where random is hard to beat, while non-contrastive models exhibit more exploitable structure.

Our contributions are:

1. We show random dimension selection is a strong baseline for post-hoc compression of contrastive embedding models (+2.2–5.0% oracle gain at dim=256), but substantially weaker for non-contrastive models (+8.2–10.1%).

2. We demonstrate that conclusions about interchangeability depend on compression ratio, importance definition, and evaluation protocol: cross-task transfer degrades at aggressive pruning (dim $\leq$ 64), Shapley-based importance reveals more structure than standalone scoring, and random selection is also more robust under domain shift.

3. We provide practical guidance via retrieval-system cost analysis and domain-shift robust-

ness experiments, establishing when random selection is sufficient and when task-aware alternatives are worth their cost.

Figure 1 summarizes our research framework: 11 embedding models across four training paradigms, eight dimension selection methods, and the key findings that emerge from their comparison.

2 Related Work

Embedding redundancy. Takeshita et al. (2025) evaluate six models on retrieval and classification, finding minor accuracy drops when randomly removing 50% of dimensions, but do not compare against task-aware selection. Tsukagoshi and Sasano (2025) find that prompt-based LLM embeddings tolerate extreme pruning on classification tasks but require more dimensions for retrieval. Zhang et al. (2024) show PCA can halve dimensions with little loss. We test whether *intelligent* selection outperforms these random baselines and show the answer depends on training paradigm.

Task-aware compression. Yoon et al. (2024) learn adaptors for efficient LLM embeddings. Faggioli et al. (2025) estimate per-dimension importance for retrieval. Zhang et al. (2025) combine Matryoshka training with adaptive selection. Anonymous (2025b) use gradient alignment for domain-specific pruning. Anonymous (2025a) propose sparse coding. These methods require model access, gradient computation, or training data. We include stronger deployable baselines (gradient saliency, activation variance, learned masks, PCA) and show that for contrastive models, even these sophisticated methods barely outperform random selection.

Matryoshka representation learning. Kusupati et al. (2023) train embeddings where the first k dimensions are optimized for any truncation level. This is a training-time intervention that requires modifying the loss function. Our sequential baseline tests whether off-the-shelf models exhibit similar properties without explicit training for it (they do not: $d < 0.02$), confirming that structured truncability must be learned, not assumed.

Neural network pruning & intrinsic dimensionality. Pruning literature (Han et al., 2016; Frankle and Carbin, 2019; Blalock et al., 2020) typically involves retraining; we prune *output embeddings* of frozen models without any weight modification. Li et al. (2018) and Alemohammad et al. (2023) show representations live on low-dimensional manifolds, suggesting compressibility. Our entropy analysis complements this by measuring *uniformity* of per-dimension importance—we find that intrinsic dimensionality alone does not predict whether task-aware selection helps, because the distribution of importance across dimensions matters as much as the total number of effective dimensions.

3 Methodology

3.1 Models and Tasks

We evaluate 11 text embedding models spanning diverse architectures and training objectives: contrastive models (GTE-Large, Stella, BGE-M3, GTE-Base, GTR-T5-Large, MxBai, Qwen3-Embedding), instruction-tuned (Instructor-Large), encoder-only (Roberta-Large, BART-Base), and task-specific fine-tuned (Roberta-InBedder). This selection covers the most widely used model families in the MTEB leaderboard and includes both contrastive and non-contrastive training paradigms. Nine models receive full chunk-level analysis (Appendix K). We use 35 MTEB tasks (Muennighoff et al., 2023) covering classification (11 tasks), clustering (3 tasks), retrieval (5 tasks), reranking (3 tasks), STS (7 tasks), pair classification (3 tasks), and summarization (1 task).

3.2 Selection Methods

Given a D-dimensional embedding $\mathbf{e} \in \mathbb{R}^D$ and target $k < D$, we compare five strategies: **(1) Random**: uniformly sample k dimensions (10 trials). **(2) Sequential**: retain the first k dimensions. **(3) Magnitude**: rank by L2 norm of embedding weight columns $\|\mathbf{w}_d\|_2$ (Han et al., 2016). **(4) Optimized (Oracle)**: divide D dimensions into chunks of size w (default $w = 2$; Appendix L), score each chunk's standalone task contribution $S_i = \mathrm{Eval}(\mathbf{e}[\mathrm{chunk}_i])$, select top chunks. This is an upper bound *within standalone chunk-scoring methods*. **(5) Anti-optimized**: select the lowest-scoring chunks (lower bound).

The first three are task-agnostic (no task-specific evaluation); the oracle bounds require full task evaluation. We also test gradient-based, activation-variance, and learned-mask methods (Section 7; Appendix M).

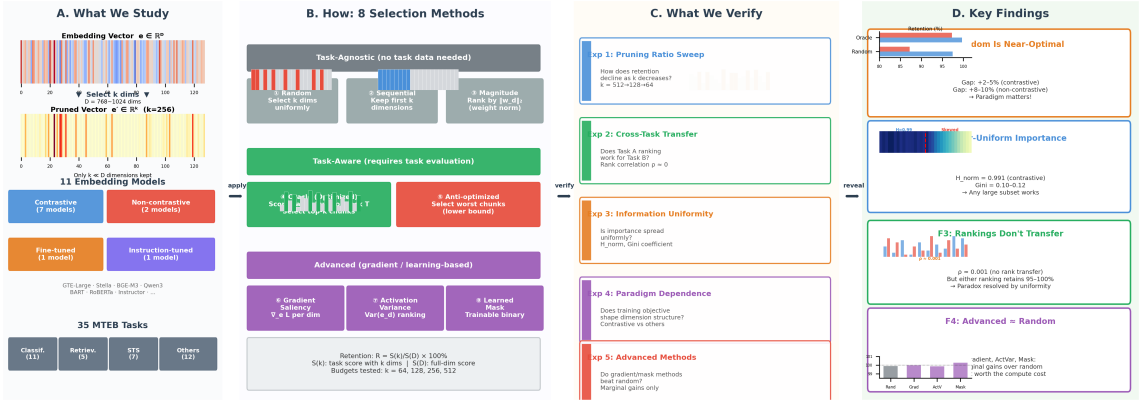

Figure 1: Research framework overview. We evaluate 8 dimension selection methods (task-agnostic, task-aware, and advanced) on 11 embedding models across 35 MTEB tasks. Key findings: (F1) random selection is near-optimal for contrastive models but not for non-contrastive ones; (F2) magnitude-based selection fails; (F3) importance rankings do not transfer across tasks but performance does; (F4) gradient-based and learned methods offer marginal improvement. Practical guidance: default to random selection (95–98% retention at 75% compression).

3.3 Evaluation Protocol

We compute the **retention ratio**: $R_k^{M,T} = S_k^{M,T}/S_D^{M,T} \times 100\%$, where $S_k^{M,T}$ is the task metric (e.g., accuracy, nDCG, Spearman ρ) for model M on task T using only the selected k dimensions. We report mean retention with 95% bootstrap CIs, paired t-tests, Wilcoxon signed-rank tests, and Holm-Bonferroni correction. We apply TOST non-inferiority tests with margin $\delta = 2\%$ to formally test whether random selection is "good enough."

4 Finding 1: Random Is Strong, but Task-Aware Gains Are Model-Dependent

Optimized selection consistently outperforms random at dim=256, but the gap varies sharply by model family (Figure 2). Table 1 shows the gap is $+2.2$–5.0% for six contrastive models (95–98% random retention) but $+8.2$–10.1% for non-contrastive models. This gradient (contrastive $<$ fine-tuned $<$ non-contrastive) reveals that dimension structure is shaped by training objective: contrastive losses explicitly distribute information across all dimensions, while reconstruction and classification objectives allow concentration.

The effect sizes tell a consistent story. Cohen's d is 0.76–1.04 for contrastive models (medium-to-large effects) but the absolute gaps remain small because baseline retention is already high (95–98%). For non-contrastive models, the gaps are both statistically significant and practically meaningful: BART-Base drops from 97.3% (random) to 87.2% (optimized)—a 10pp gap that would materially affect downstream performance.

Table 1: Oracle vs. random at dim=256. Δ: mean retention gap with 95% CI; d: Cohen's d; p_{holm}: Holm-Bonferroni corrected.

Model	Δ	95% CI	d	p_{holm}
Contrastive models				
Instructor-Large	+2.89%	[+1.78, +4.09]	0.83	<0.001
MxBai-Embed-Large	+2.24%	[+1.35, +3.29]	0.76	<0.001
GTE-Large	+2.41%	[+1.49, +3.39]	0.83	<0.001
Qwen3-Embedding	+5.01%	[+3.36, +6.84]	0.97	<0.001
Stella EN 400M	+3.22%	[+2.17, +4.44]	0.93	<0.001
BGE-M3	+4.62%	[+3.21, +6.13]	1.04	<0.001
Non-contrastive / fine-tuned				
Roberta-InBedder	+8.19%	[+5.99, +10.56]	1.17	<0.001
BART-Base	+10.08%	[+6.28, +13.73]	0.90	<0.001
Roberta-Large	+8.56%	[−5.85, +18.88]	0.23	0.191

TOST non-inferiority confirms paradigm split. Pooling contrastive models, the mean gap is -0.83% (non-inferiority established: $p_{\text{TOST}} = 0.0009$ at $\delta = 2\%$). For non-contrastive models, the gap is -3.41% (non-inferiority fails: $p_{\text{TOST}} = 0.96$). The split holds across margins $\delta \geq 2$pp (Appendix A). This formal statistical test confirms that random selection is not merely "close to" optimized selection for contrastive models—it is provably non-inferior within a 2pp margin.

Sequential $\approx$ Random ($|d| < 0.02$). The intrinsic dimension ordering carries no importance signal—off-the-shelf models do not exhibit Matryoshka-like structure. This contrasts with models explicitly trained with Matryoshka Representation Learning (Kusupati et al., 2023), where the first k dimensions are specifically optimized to be informative. The absence of such structure in standard models confirms that truncation-ordering

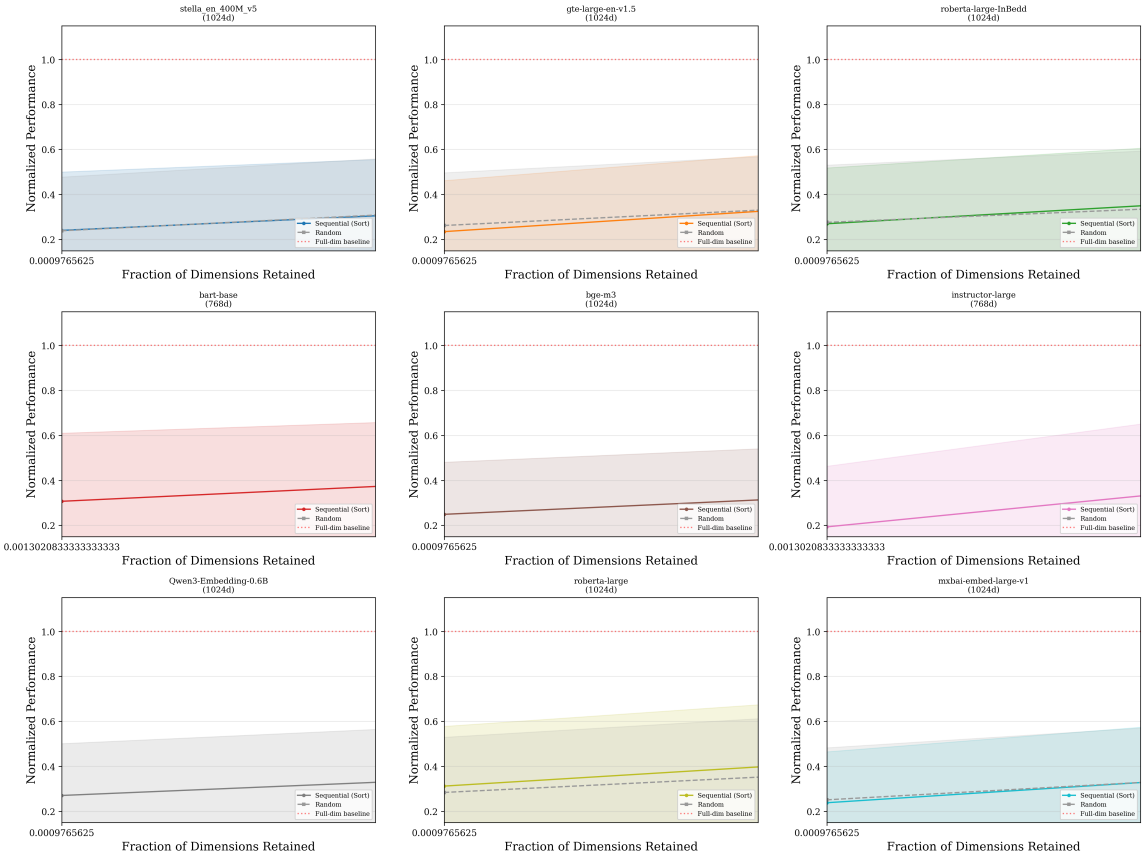

Figure 2: Retention vs. pruning ratio. The Optimized–Random gap is small for contrastive models (GTE-Large, Stella) but large for non-contrastive models (BART-Base, Roberta-Large). Sequential closely tracks Random.

is a product of the training objective, not an inherent property of embedding spaces.

Anti-optimized confirms information exists. Despite the interchangeability of "good" methods, anti-optimized degrades to 84–91% (Figure 3). Information is distributed but not absent—it is spread so evenly that subset choice barely matters at moderate compression. The existence of this large anti-optimized gap (10–16pp below random) also establishes an important boundary condition: were the information truly concentrated in a small subset, anti-optimized selection would not degrade this much.

5 Finding 2: Magnitude-Based Selection Fails

Magnitude rankings are uncorrelated with task importance ($\rho \approx 0$; Figure 4(a)). For GTE-Large, magnitude is *significantly worse* than random (96.74% vs. 97.49%, $p = 0.002$, $d = -0.27$). The top-128 chunks by magnitude share only 18.8% with top-128 by task importance (Jaccard $= 0.10$): magnitude measures encoding activation, not downstream utility. This result is robust across model families—for every model tested, magnitude fails to outperform random selection, and for non-contrastive models (Roberta-Large, BART-Base) the degradation is even more severe. The failure of magnitude-based selection is particularly notable because it is the default pruning heuristic in the neural network compression literature (Han et al., 2016), suggesting that transfer of pruning intuition from weights to embeddings requires care.

6 Finding 3: Cross-Task Transfer Paradox

At dim=256, using any task's importance ranking to prune for any other task retains 94.8–100.4% across 11 models (Table 2), despite near-zero rank correlation ($\rho \approx 0.001$). This paradox is resolved by near-uniform importance: when all dimensions contribute roughly equally, any sufficiently large subset works (Figure 5). The paradox has a practical implication: practitioners do not need task-specific importance rankings for each downstream application. A single ranking computed on any reasonable proxy task suffices.

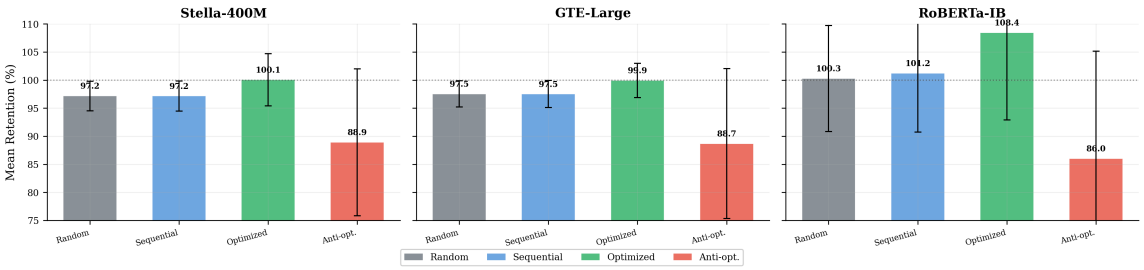

Figure 3: All five methods at dim=256. Random, Sequential, and Magnitude cluster together (96–98%); Anti-optimized drops to 84–91%.

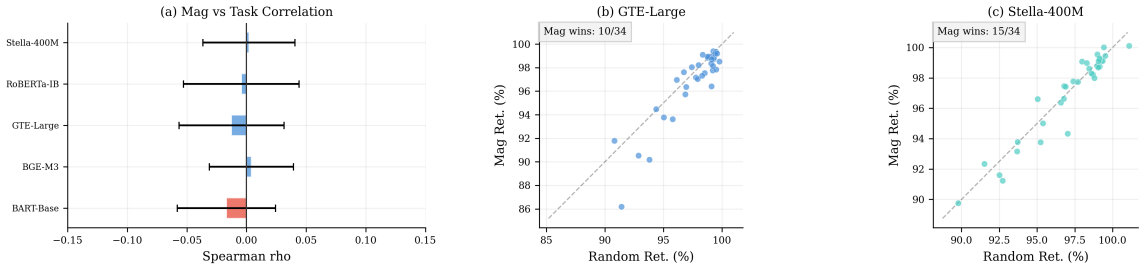

Figure 4: (a) Magnitude–task importance correlation is near-zero across models. (b–c) Per-task magnitude vs. random retention; most points fall below the diagonal for GTE-Large.

Table 2: Cross-task transfer at dim=256: near-complete retention despite zero ranking correlation.

Model	Mean Transfer Ret.	95% CI
Roberta-Large	100.4%	[87.9, 123.8]
Roberta-InBedder	99.2%	[91.2, 109.4]
GTE-Large	97.5%	[90.7, 100.1]
Stella	97.4%	[91.2, 100.7]
MxBai-Embed-Large	97.2%	[91.3, 99.9]
Qwen3-Embedding	96.8%	[96.6, 97.1]
BGE-M3	94.8%	[81.2, 101.1]

Table 3: Mean retention (%) at budget=256. Gradient and activation variance can be actively harmful for non-contrastive models.

Model	Random	Gradient	Act. Var.	Mask
GTE-Large	99.5	99.5	99.7	98.9
Stella	98.8	99.3	98.4	98.3
BGE-M3	96.8	97.1	97.6	96.4
Roberta-Large	102.9	93.4	92.4	102.0
InBedder	100.1	94.8	93.3	98.5

Transfer degrades sharply at aggressive compression: at $k = 16$, retention drops to 44–50% for contrastive and 33–35% for non-contrastive models (Appendix B). Cross-task transfer is viable at dim ≥ 128, but task-specific estimation becomes necessary below that. This degradation pattern mirrors the main pruning results: at very aggressive compression, the "any large subset works" property breaks down because the retained subset is no longer large enough to capture the broadly distributed information.

7 Finding 4: Stronger Baselines Also Fail

We test gradient-based importance, activation variance, and learned masks on 5 models (Table 3; Appendix M). For contrastive models, gains are marginal (+0.0 to +0.8%). For non-contrastive models, gradient and activation-variance methods are *actively harmful*: Roberta-Large drops from 102.9% (random) to 93.4% (gradient). Only the oracle reliably beats random, but it requires full task evaluation.

The failure of gradient-based methods for non-contrastive models is particularly striking. Gradient saliency identifies dimensions that are important for *loss minimization* during training, but these dimensions do not align with importance for downstream task performance in the compressed embedding space. Similarly, activation variance captures which dimensions are most variable during inference, which does not translate to downstream utility. The learned mask approach, which directly optimizes a binary mask on the embedding dimensions, performs better for non-contrastive models (Roberta-Large: 102.0%, InBedder: 98.5%) but still does not consistently beat random for contrastive models.

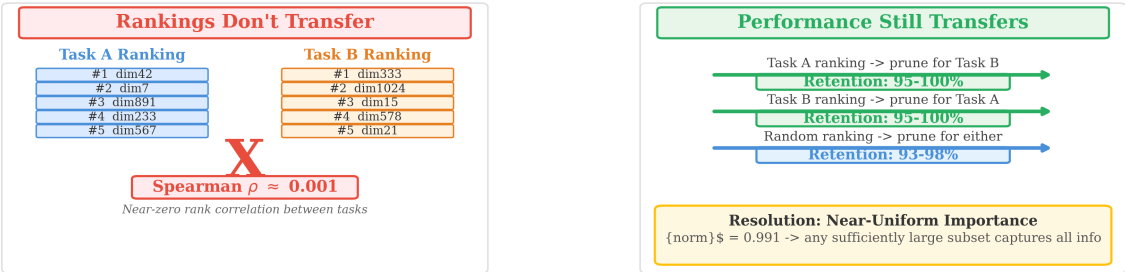

Figure 5: Cross-task transfer paradox. Rankings disagree ($\rho \approx 0.001$), yet either ranking retains 95–100% at dim=256. Resolution: near-uniform importance ($H_{\text{norm}} = 0.991$) means any large subset works.

8 Analysis

Near-uniform importance. We compute normalized Shannon entropy of chunk importance: $H_{\text{norm}} = -\frac{1}{\log_2 N} \sum_i p_i \log_2 p_i$ where $p_i = S_i / \sum_j S_j$. Values are near-maximum (contrastive: 0.993–0.994; non-contrastive: 0.989), confirming broadly distributed importance (Figure 6). However, Shapley-based entropy (0.60–0.66) is much lower than standalone entropy (0.997–0.999), revealing that interaction-aware importance detects substantially more structure (Appendix F).

Training paradigm predicts the gap. Linear regression (gap $\sim$ contrastive + params, $n = 173$) shows the contrastive indicator is significant ($\beta = -2.77$, $p < 0.001$) while model size is not ($p = 0.37$). The paradigm gap is largest for STS tasks (1.26% vs. 6.05%) and smallest for classification (1.76% vs. 4.05%). Table 4 shows non-contrastive models have lower entropy and higher Gini coefficients, providing a quantitative explanation: more concentrated importance distributions create more room for task-aware selection to exploit.

Compression ratio matters. The oracle–random gap grows with compression severity. At dim=256 (75% reduction), the gap is +2–5% for contrastive models. At dim=64 (94% reduction), it widens to +6–12%. At dim=16, random retention drops to 44–63%, making any form of coordinate selection impractical. This scaling pattern suggests that random selection is adequate for moderate compression but task-aware methods become increasingly valuable as the budget shrinks. The transition point differs by model family: contrastive models tolerate more aggressive random pruning before the gap becomes practically significant.

Table 4: Importance distribution metrics (win_size=2). Non-contrastive models show lower entropy, higher Gini, larger oracle gaps.

Metric	GTE-Large	Stella	InBedder	BART	Roberta
Entropy	0.994	0.993	0.990	0.989	0.989
Gini	0.103	0.108	0.124	0.132	0.133
Opt.–random gap	+2.58%	+1.90%	+3.71%	+5.01%	+6.20%

Table 5: Retrieval cost–quality tradeoff at dim=256 (GTE-Large). Random achieves 78–97% nDCG with 4× memory reduction.

Task	Method	nDCG10 (%)	Mem (MB)	Latency (ms)	Recall (%)
ArguAna	Full-1024	100.0	19.5	0.21	100.0
	Random-256	97.3	4.9	0.06	96.3
	RP-256	98.7	4.9	0.39	95.1
SciFact	Full-1024	100.0	19.5	0.22	100.0
	Random-256	87.3	4.9	0.06	90.1
	RP-256	84.7	4.9	0.40	88.7

Retrieval system cost. Random-256 achieves 78–97% of full-dim nDCG with 4× memory reduction on FAISS (Table 5). PCA catastrophically fails for retrieval (near-zero nDCG) under default indexing; random projection avoids this but requires a projection matrix. Notably, random-256 reduces P50 latency from 0.21–1.82ms to 0.04–0.06ms, a 4–30× speedup, while maintaining 78–97% of retrieval quality.

Coordinate selection vs. PCA and random projection. Table 6 compares coordinate selection (RC) against PCA and random projection (RP). At dim=256, all methods converge for contrastive models (95–99% retention). At dim=64, RC excels for non-contrastive models (Roberta-Large: 98.4% vs. PCA 80.6%), because PCA's global rotation destroys the coordinate-aligned structure that non-contrastive models exploit. This suggests coordinate selection as a practical middle ground: zero-cost like random, compatible with existing indexing, and competitive with projection methods at moderate compression.

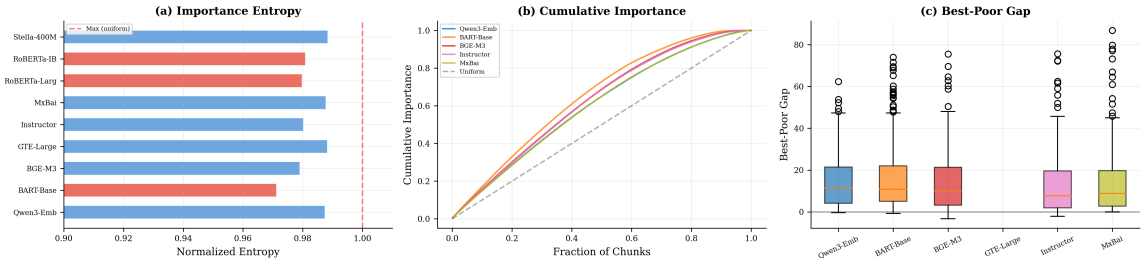

Figure 6: (a) Normalized entropy near 1.0 for all models. (b) Cumulative importance rises gradually: top 50% of chunks capture only 58–75%.

Table 6: Coordinate selection (RC) vs. PCA and random projection (RP). At dim=256, methods converge; at dim=64, RC excels for non-contrastive models.

Model	Dim	PCA	RP	RC
GTE-Large	64 / 256	90.2 / 98.0	92.9 / 98.7	93.1 / 99.0
Stella	64 / 256	92.0 / 98.5	93.9 / 98.6	94.4 / 98.9
BGE-M3	64 / 256	85.9 / 95.8	87.2 / 96.4	87.8 / 96.1
Roberta-L.	64 / 256	80.6 / 99.9	91.2 / 99.6	98.4 / 100.2
InBedder	64 / 256	80.8 / 95.6	90.4 / 98.4	96.7 / 98.0

Table 7: OOD transfer gap at dim=256. Positive = source-optimized helps. For contrastive models, OOD selection $\approx$ random.

Pair	GTE	Stella	InBedder	BGE-M3
NFCorpus $\rightarrow$ ArguAna	+0.4	-0.8	-7.9	-0.6
NFCorpus $\rightarrow$ SciFact	+0.6	+1.8	-3.4	-2.7
IMDb $\rightarrow$ Banking77	-0.4	+1.1	+0.8	-0.9
STSB $\rightarrow$ BIOSSES	+0.3	-0.0	+8.5	-1.9

Robustness. Random selection has manageable tail risk at dim $\geq$ 128 (P5 is 1–3% below mean for contrastive models; Appendix Q). At dim=16, variance increases sharply (CV: 4.5–9.7%), confirming that the "any subset works" property breaks down at extreme compression. Under domain shift (Table 7), random is more robust than task-aware selection for contrastive models, avoiding negative transfer—OOD-optimized selection produces gaps of -0.8 to $+1.8$pp, comparable to random. For non-contrastive models, domain transfer can help or hurt depending on task similarity.

9 Discussion

For *contrastive models*, random coordinate selection retains 95–98% at 75% compression, making it a strong zero-cost baseline. For *non-contrastive models*, task-aware selection provides meaningful gains (+8–10%). At dim=256, coordinate selection matches PCA and random projection for contrastive models, while preserving compatibility with existing retrieval infrastructure (Appendix G). Magnitude-based pruning should be avoided regardless of model type.

Why does random work? The near-uniform entropy ($H_{\text{norm}} = 0.989$–0.994) means that each dimension (or chunk) contributes a roughly equal share of task performance. This is a direct consequence of contrastive training, which distributes semantic information across the full embedding space to maximize representational capacity. Non-contrastive models, trained with reconstruction or classification objectives, concentrate information more unevenly, creating exploitable structure for task-aware pruning.

When is random not enough? Three regimes demand task-aware selection: (1) non-contrastive models, where the oracle–random gap is +8–10%; (2) aggressive compression (dim $\leq$ 64), where random retention drops below 93% and variance increases sharply; and (3) domain shift scenarios where the target domain is unknown at compression time, though in this case even oracle selection struggles.

Relation to Matryoshka representations. Our sequential baseline tests whether off-the-shelf models exhibit Matryoshka-like structure. The answer is definitively no ($|d| < 0.02$ for sequential vs. random), confirming that the structured information distribution in Matryoshka-trained models (Kusupati et al., 2023) is a product of the training objective, not an emergent property of embedding spaces.

Limitations. (1) Our oracle is an upper bound *within standalone chunk-scoring methods*, not universally. Greedy forward selection slightly outperforms contiguous selection (Appendix O), but the gap is modest. (2) Conclusions are strongest at moderate compression; transfer degrades at dim $\leq$ 64. (3) The training-paradigm association is obser-

vational, not causal—we cannot rule out confounds from model size, data, or other architectural differences. (4) We evaluate on MTEB tasks only; results may differ for other embedding use cases (e.g., few-shot prompting, RAG).

10 Conclusion

Random dimension selection is a strong baseline for post-hoc compression of contrastive embedding models (+2.2–5.0% oracle gap at dim=256), but the gap is substantially larger for non-contrastive models (+8.2–10.1%). The near-uniform importance we observe (entropy $= 0.989$–0.994) explains why random works, but Shapley-based analysis (entropy $= 0.60$–0.66) shows this is partly a consequence of standalone scoring. These findings establish a practical rule: for contrastive models at moderate compression, random coordinate selection is the recommended default; for non-contrastive models or aggressive compression, task-aware selection provides meaningful gains. Future work should investigate whether these findings extend to multimodal embeddings and larger-scale retrieval benchmarks, and explore training-time interventions that preserve compression-friendly structure. Code and results will be released upon publication.

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A Detailed TOST Analysis

Table 8 shows TOST non-inferiority results across margins. Contrastive models pass at all $\delta \geq 2$pp ($p < 0.001$), while non-contrastive models fail up to 4pp.

Table 8: TOST sensitivity across margins.

δ (pp)	Contrastive	Non-contrastive	Verdict
1	0.13	>0.99	Both fail
2	<0.001	0.98	Split
3	<0.001	0.79	Split
4	<0.001	0.28	Split
5	<0.001	0.03	Both pass

B Transfer Degradation Details

Table 9: Mean cross-task transfer retention (%) across compression levels ($w = 4$, 13 tasks). Transfer degrades sharply below dim=64.

Model	k=16	k=32	k=64	k=128	k=256
GTE-Large	48.7	55.1	60.3	62.6	63.6
Stella	50.2	56.4	61.9	64.6	65.5
BGE-M3	43.8	49.6	55.2	58.3	59.4
Roberta-Large	35.2	38.6	42.8	44.4	45.9
InBedder	33.3	35.6	38.1	40.1	40.2

C All Methods Comparison

D Magnitude-Based Selection Details

E Redundancy Mechanism

F Importance Definition Comparison

Table 10: Spearman correlation between standalone and alternative importance definitions ($w = 4$, budget=256), and entropy per definition.

	Correlation with Standalone			Entropy			
Model	LOO	Marg.	Shap.	Stand.	LOO	Marg.	Shap.
GTE-Large	0.094	0.155	0.132	0.999	0.937	0.939	0.597
Stella	0.124	0.209	0.157	0.999	0.918	0.944	0.610
Roberta-Large	0.091	0.296	0.235	0.997	0.838	0.891	0.660
InBedder	0.096	0.248	0.179	0.997	0.888	0.927	0.654

G Retrieval System Cost Details

See Table 5 in the main body for the full retrieval cost–quality tradeoff analysis. The main body table shows three representative retrieval tasks; additional tasks show consistent patterns.

H PCA and Random Projection Comparison

See Table 6 in the main body for the coordinate selection vs. PCA and random projection comparison.

I Per-Task Results for GTE-Large

J Detailed Statistical Tests

Table 12 provides detailed statistical comparisons between all method pairs at dim=256.

K Model List

Table 13 lists all 11 models evaluated in the study.

L Chunk Size Sensitivity

We use chunk size $w = 2$ throughout our main experiments. To verify that our conclusions are not sensitive to this choice, we sweep chunk sizes $w \in \{1, 2, 4, 8, 16, 32\}$ on 5 models across 6 representative tasks at budgets 128 and 256.

Table 14 shows selected results. The key findings are: (1) for contrastive models, the optimized–random gap remains small across all chunk sizes at moderate compression, confirming that no granularity level reveals exploitable structure; (2) for non-contrastive models, the gap is consistently large regardless of chunk size; (3) there is no monotonic relationship between chunk size and gap magnitude—$w = 1$ does not systematically inflate or deflate the gap. Normalized entropy remains near-maximum across all chunk sizes ($H_{\text{norm}} = 0.997$–0.999). These results validate our choice of $w = 2$ and confirm that the training-paradigm-dependent gap is a robust finding across aggregation levels.

M Gradient and Saliency-Based Baselines

Table 15 shows detailed per-task results for gradient-based, activation variance, and learned mask selection on all 5 evaluated models at budget=256.

N Leave-One-Out Importance Analysis

Table 16 shows per-task correlations and entropy for all four importance definitions on Roberta-Large.

The weak correlation between standalone and LOO/marginal/Shapley importance confirms that chunk importance is inherently context-dependent. Standalone entropy is near-maximum (0.985–1.000) while Shapley entropy is substantially lower (0.655–0.933), revealing that interaction-aware importance detects much more structure. For STS-Benchmark, LOO entropy drops to 0.343—far from uniform—indicating that some chunks are crucial in context even though they appear unimportant in isolation.

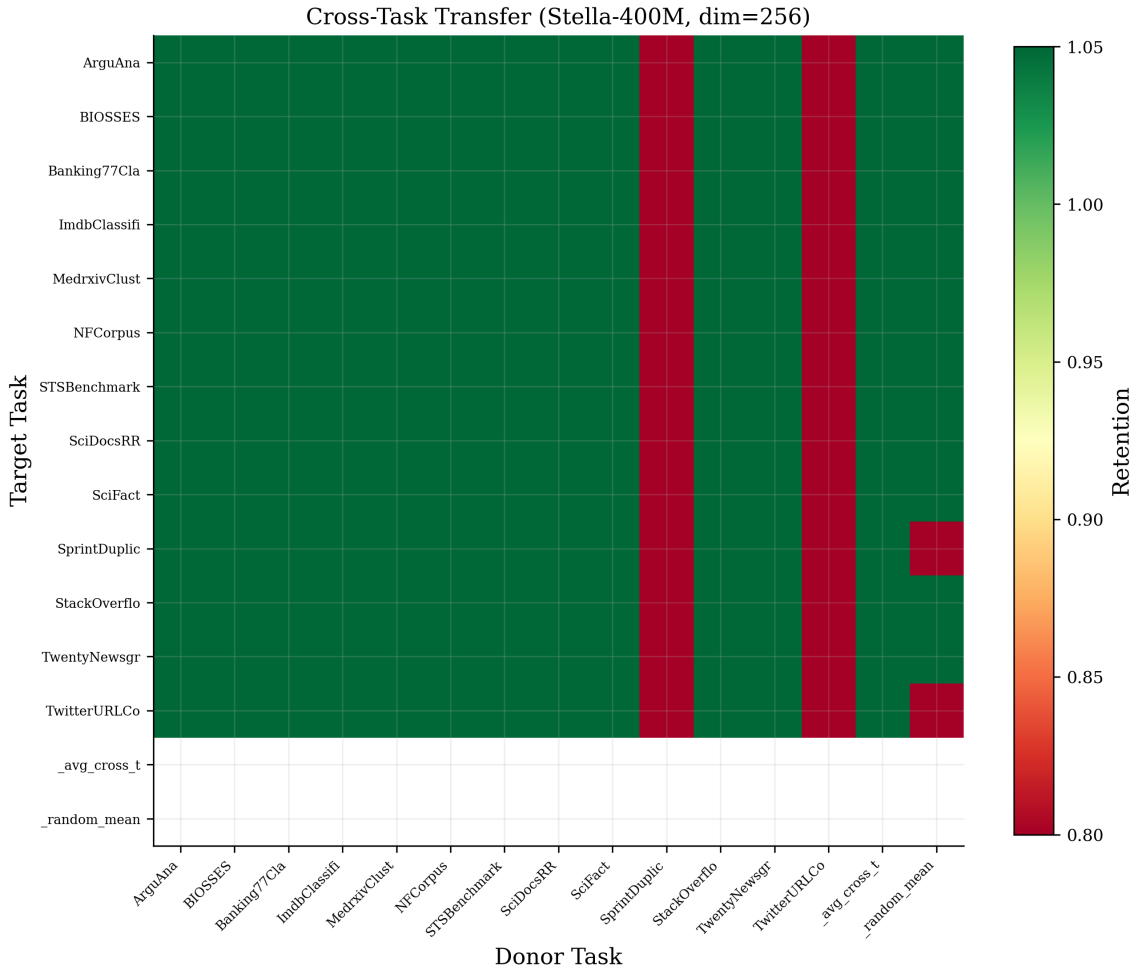


Figure 7: Cross-task transfer heatmap for a representative model (GTE-Large). Each cell shows retention when using the source task's ranking to prune for the target task. Near-diagonal entries (self-transfer) are high; off-diagonal entries are also high despite low rank correlation.

O Non-Contiguous Selection

Our oracle selects contiguous chunks. A concern is whether non-contiguous (scattered) dimension selection could achieve better performance.

Greedy forward selection slightly outperforms the contiguous oracle on some tasks (Table 17), but the differences are modest for contrastive models. The Jaccard overlap between oracle and greedy selections is very low (0.07–0.19), meaning the two methods select fundamentally different dimension subsets—yet achieve similar performance. This confirms that the *specific* dimensions chosen matter less than the *number* of dimensions retained, validating our use of contiguous selection in the main analysis.

P Out-of-Domain Robustness

Q Random Variance and Tail Risk

We characterize the full distribution of random selection performance using 100 seeds per model-task-budget combination across 9 models and 5 budgets ($k \in \{16, 32, 64, 128, 256\}$).

Table 19 summarizes distribution statistics. For contrastive models at dim=256, the coefficient of variation is $<1\%$, and the 5th percentile is within 1pp of the mean. Even at aggressive compression (dim=16), where CV rises to 4–10%, the CVaR-5 remains within 5pp of the mean, indicating no catastrophic outliers.

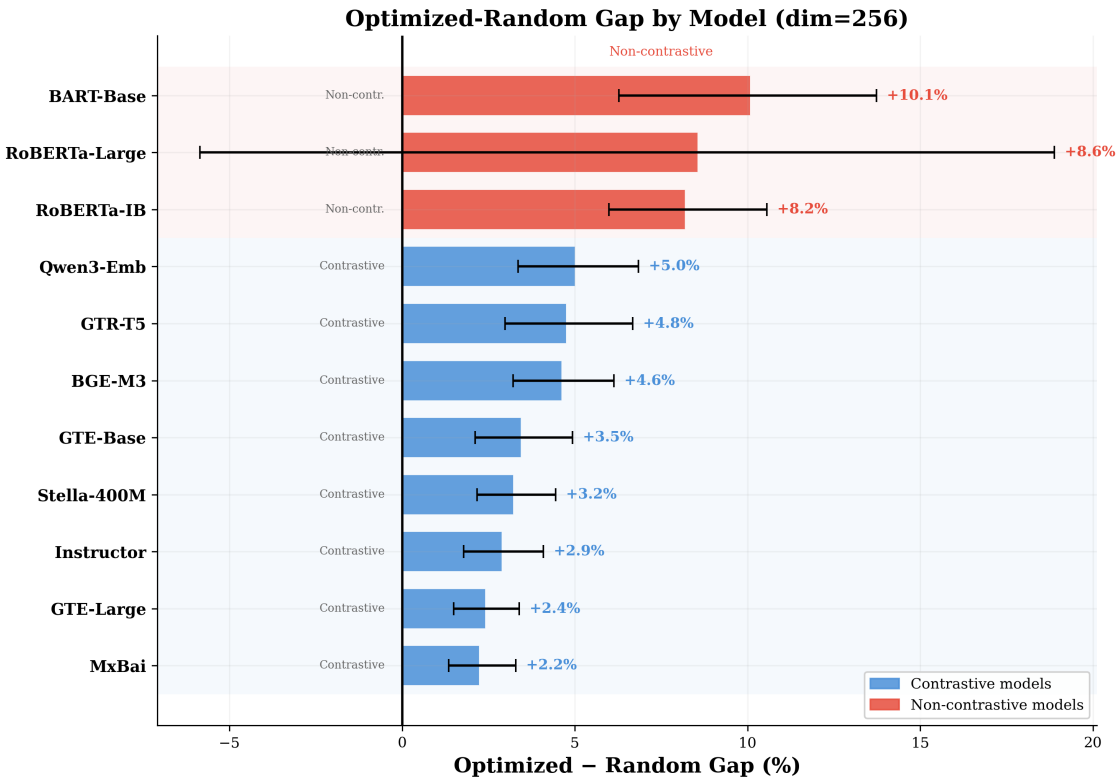


Figure 8: Optimized–random gap for all 11 models. Contrastive models (blue) cluster around +2–5%, while non-contrastive models (red) show +8–10% gaps. Error bars show 95% CI.

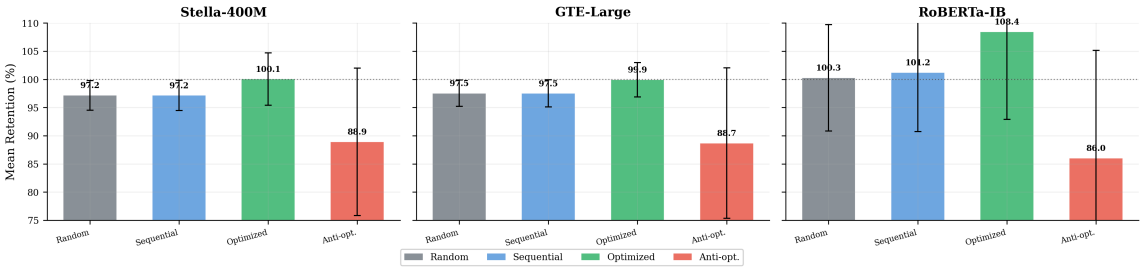


Figure 9: All five methods at dim=256. Random, Sequential, and Magnitude cluster together (96–98%); Anti-optimized drops to 84–91%.

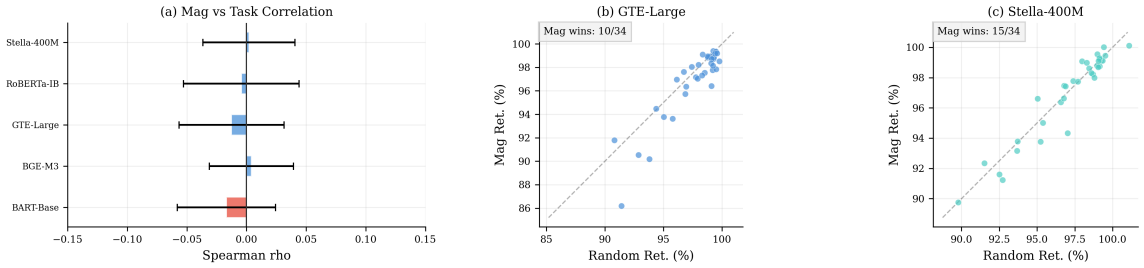


Figure 10: (a) Magnitude–task importance correlation is near-zero across models. (b–c) Per-task magnitude vs. random retention; most points fall below the diagonal for GTE-Large.

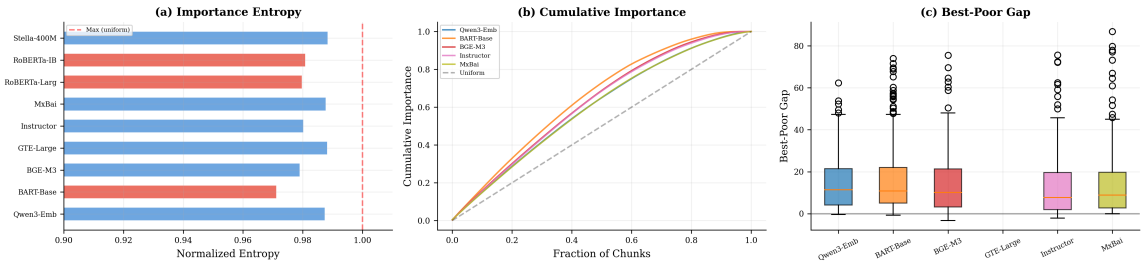


Figure 11: (a) Normalized entropy near 1.0 for all models. (b) Cumulative importance rises gradually: top 50% of chunks capture only 58–75%.

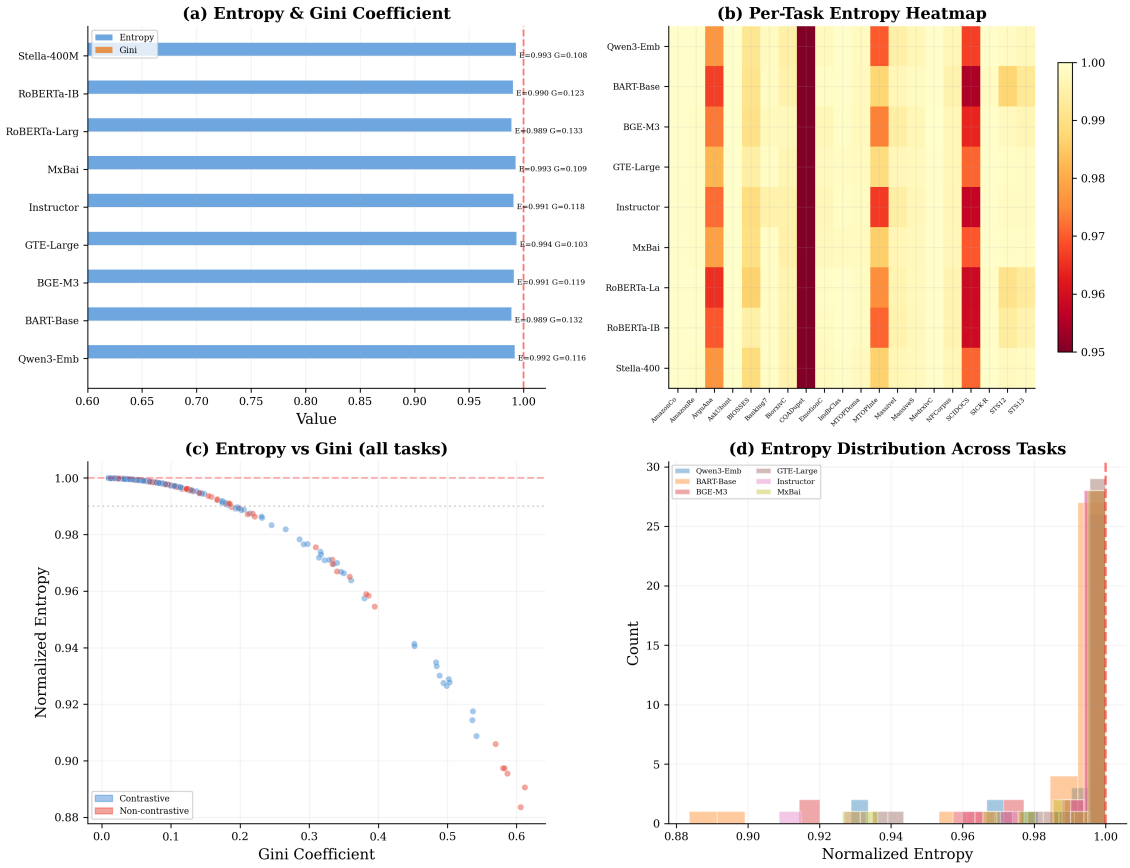


Figure 12: Entropy and Gini coefficient for all 9 core models across MTEB tasks. All contrastive models show $H_{\text{norm}} > 0.985$, confirming near-uniform dimension importance.

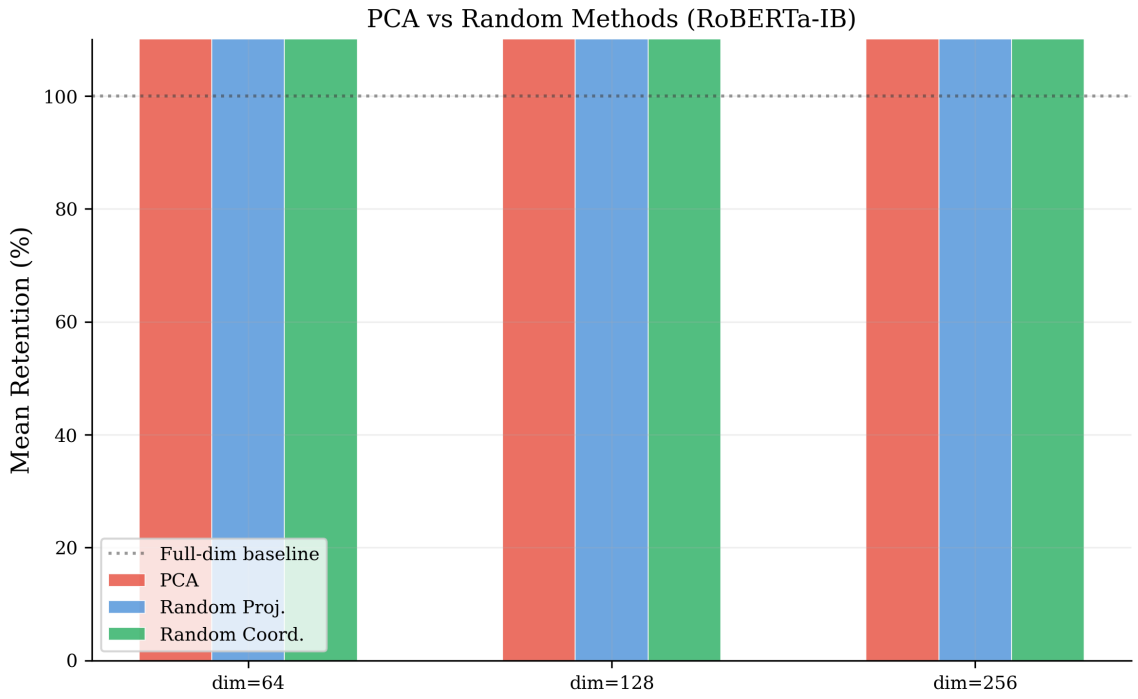


Figure 13: PCA vs. random projection vs. coordinate selection at dim=64/128/256. PCA's global rotation destroys coordinate-aligned structure, causing catastrophic degradation for contrastive models. Coordinate selection preserves infrastructure compatibility while matching random projection quality.

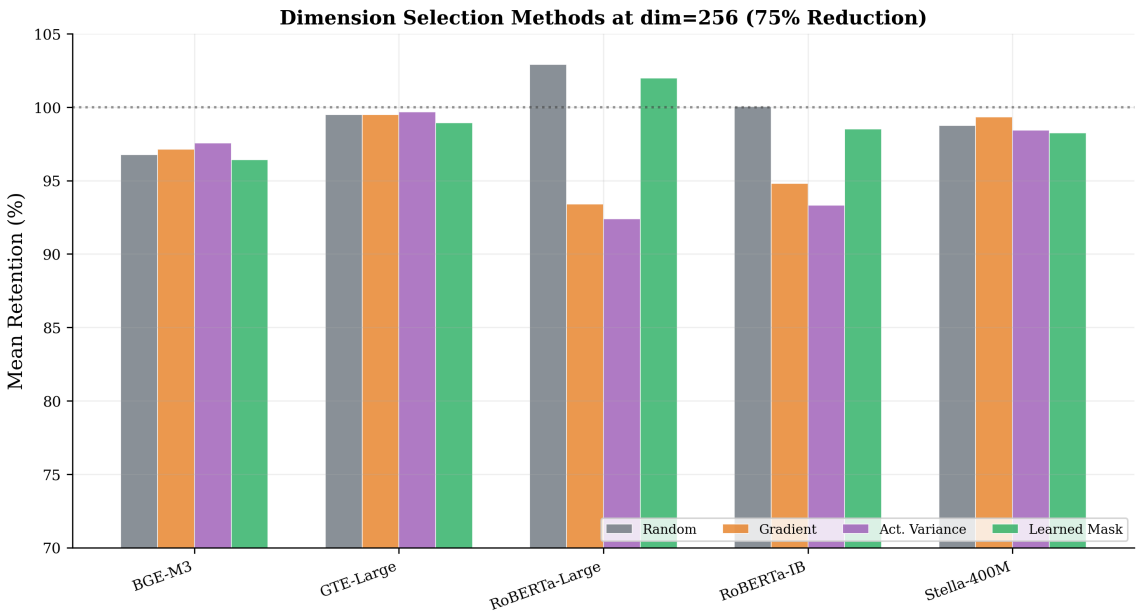


Figure 14: Grouped bar chart comparing Random, Gradient, Activation Variance, and Learned Mask selection at dim=256. Gradient-based and activation-based methods fail to consistently outperform random selection, particularly for contrastive models.

Table 11: Per-task retention at dim=256 (75% reduction) for all methods on GTE-Large. Random, Sequential, Optimized, and Anti-optimized use chunk-based selection (win_size=2, top/bottom 128 of 512 chunks). Magnitude uses L2-norm ranking of embedding weights. 34 of 35 evaluated tasks shown; STS17 excluded due to unreliable baseline.

Task	Random	Sequential	Magnitude	Optimized	Anti-opt.
Classification					
AmazonCntrfCls	95.8	97.3	93.6	105.6	70.0
AmazonReviewsCls	95.0	95.7	93.8	103.9	69.4
Banking77Cls	99.3	99.4	99.4	100.0	98.1
EmotionCls	94.4	93.9	94.5	100.9	82.0
ImdbCls	99.3	100.3	98.7	100.9	77.5
MTOPDomainCls	99.0	99.0	99.0	99.9	95.5
MTOPIntentCls	97.8	97.5	97.2	99.2	94.7
MassiveIntentCls	98.0	98.6	98.2	99.5	96.7
MassiveScenarioCls	97.4	97.6	98.0	99.8	91.2
ToxicConvCls	96.9	98.5	96.4	105.0	66.6
TweetSntExCls	97.9	98.0	97.0	101.2	84.8
Clustering					
BiorxivClustS2S	96.1	96.7	97.0	99.9	85.0
MedrxivClustS2S	98.5	97.4	97.5	101.0	95.6
TwentyNewsgrpClust	90.8	90.8	91.8	97.3	63.9
Retrieval					
ArguAna	96.7	96.4	97.6	96.7	96.7
CQADupstackEngRetr	93.8	93.3	90.2	90.5	94.4
NFCorpus	91.4	91.6	86.2	93.9	89.3
SCIDOCS	92.9	92.6	90.5	93.0	89.7
SciFact	96.9	95.1	95.7	97.6	96.3
Reranking					
AskUbuntuDupQ	99.2	97.7	97.8	102.2	96.6
SciDocsRR	98.3	98.6	97.3	99.4	94.7
StackOverflowDupQ	98.8	98.6	98.8	99.6	96.7
STS					
BIOSSES	98.8	99.1	98.9	105.1	94.0
SICK-R	99.5	99.4	97.9	99.8	97.1
STS12	99.6	99.1	99.2	103.4	96.9
STS13	98.3	98.7	99.1	98.3	94.4
STS14	99.1	99.7	98.7	99.5	97.6
STS15	99.1	98.9	98.4	100.2	92.5
STS16	99.1	98.6	96.4	101.7	94.0
STSBenchmark	99.2	99.7	98.2	100.4	97.6
Pair Classification					
SprintDupQ	99.5	99.4	99.4	100.9	97.0
TwitterSemEval2015	99.3	98.9	98.9	101.7	96.4
TwitterURLCorpus	99.4	99.3	99.2	100.0	98.3
Summarization					
SummEval	99.8	100.4	98.5	98.7	98.8
Mean	97.5	97.5	96.7	99.9	90.6

Table 12: Detailed statistical comparisons at dim=256. Cohen's d effect sizes (paired), paired t-test p-values, and Holm-Bonferroni corrected values.

Model	Comparison	Δ Mean	95% CI	d	t/W p	p_{holm}
Stella	Seq vs Rnd	-0.02%	$[-0.34\%, +0.36\%]$	-0.006	0.925	1.000
Stella	Mag vs Rnd	-0.12%	$[-0.39\%, +0.13\%]$	-0.045	0.377	1.000
GTE-Large	Seq vs Rnd	$+0.03\%$	$[-0.20\%, +0.26\%]$	$+0.011$	0.820	1.000
GTE-Large	Mag vs Rnd	-0.76%	$[-1.22\%, -0.35\%]$	-0.278	0.002	0.010*
Roberta-InBedder	Seq vs Rnd	$+0.83\%$	$[+0.02\%, +1.59\%]$	$+0.226$	0.049	0.196

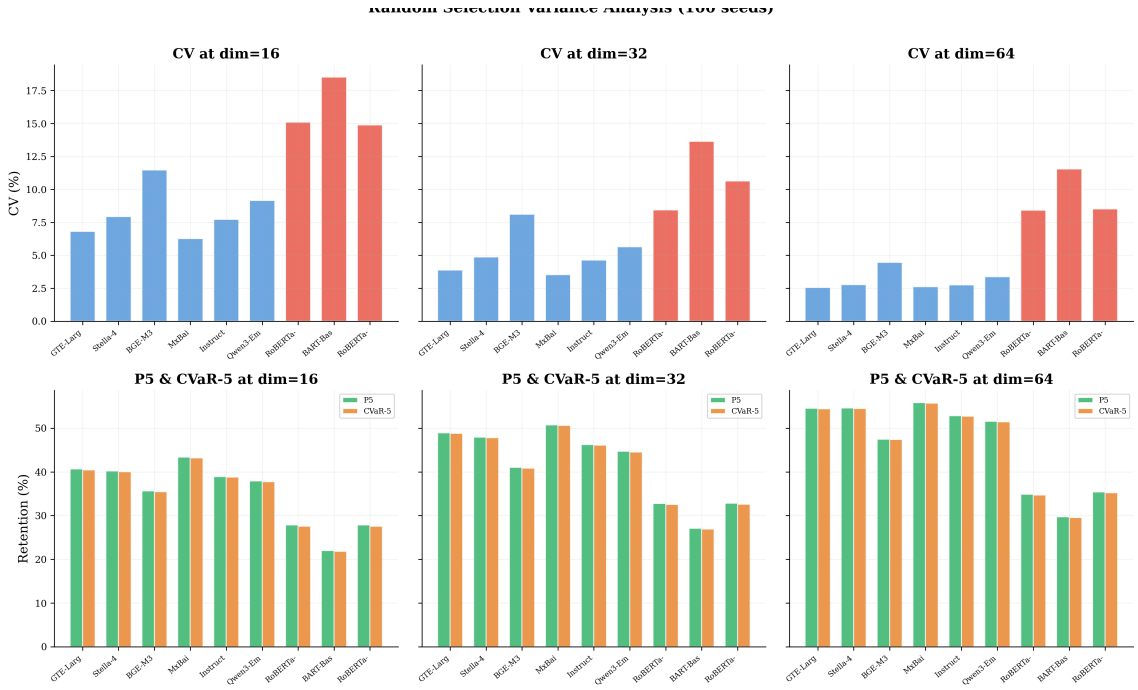
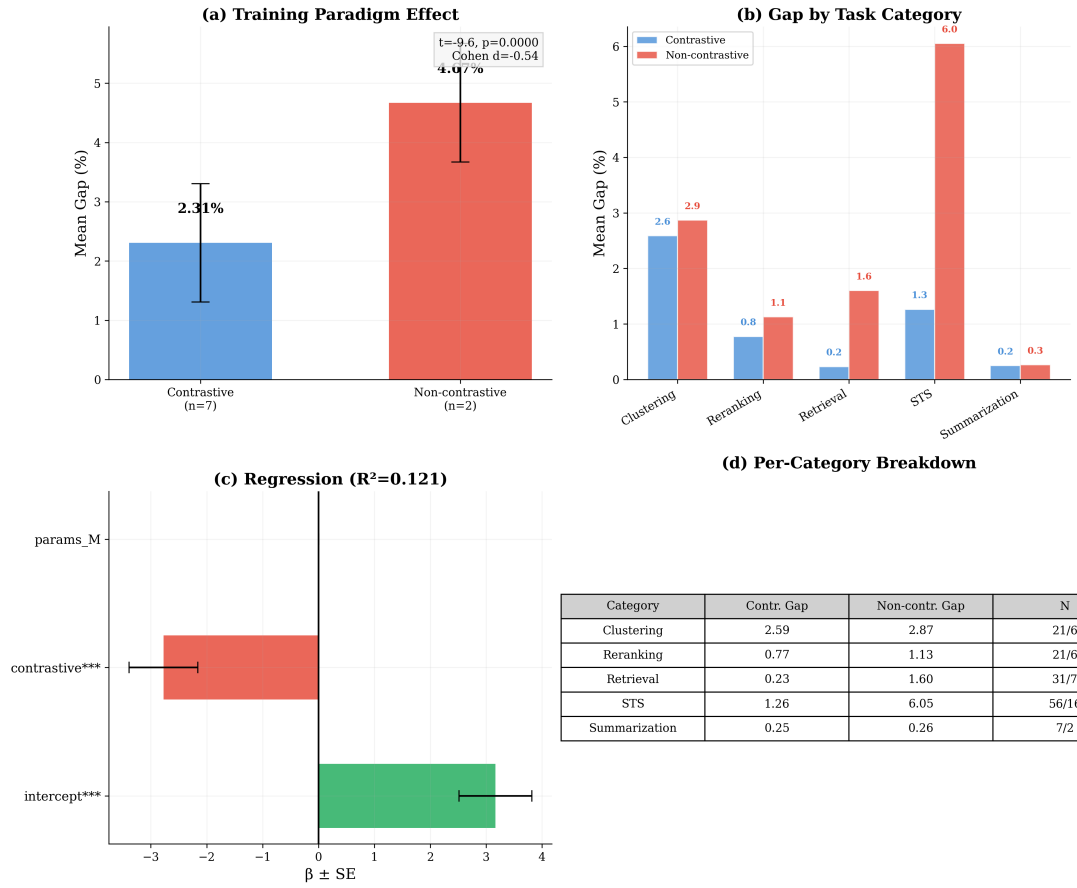


Figure 15: Distribution of random selection retention across 100 seeds for representative models. Violin plots show the full distribution shape; box plots mark quartiles. Variance increases at aggressive compression (dim ≤ 32) but the mean remains stable.



Category	Contr. Gap	Non-contr. Gap	N
Clustering	2.59	2.87	21/6
Reranking	0.77	1.13	21/6
Retrieval	0.23	1.60	31/7
STS	1.26	6.05	56/16
Summarization	0.25	0.26	7/2

Figure 16: Contrastive vs. non-contrastive training paradigm analysis. (a) Mean optimized–random gap by paradigm with Cohen's d effect size. (b) Per-task-type gap comparison showing the paradigm effect is consistent across task categories.

Table 13: Models evaluated in this study. Checkmarks indicate available data. Nine models have full chunk-level analysis.

Model	Dims	Gap	Transfer	Sweep	Mechanism
GTE-Large	1024	✓	✓	✓	✓
Stella EN 400M	1024	✓	✓	✓	✓
Roberta-InBedder	1024	✓	✓	✓	✓
BGE-M3	1024	✓	✓	✓	✓
Instructor-Large	768	✓	✓	✓	✓
MxBai-Embed-Large	1024	✓	✓	✓	✓
Qwen3-Embedding	1024	✓	✓	✓	✓
Roberta-Large	1024	✓	✓	✓	✓
BART-Base	768	✓	✓	✓	✓
GTE-Base	768		✓		
GTR-T5-Large	768		✓		

Table 14: Mean optimized–random gap (%) across 6 tasks for different chunk sizes and budgets. The gap is stable across chunk sizes for contrastive models and consistently larger for non-contrastive models. No monotonic relationship between chunk size and gap magnitude exists.

		Budget=128			Budget=256		
Model	Task	w=1	w=2	w=4	w=1	w=2	w=4
Contrastive models							
GTE-Large	IMDb	0.12	0.80	0.32	0.06	0.25	0.40
GTE-Large	STS	−0.43	1.67	1.74	−0.09	1.44	1.05
GTE-Large	TwentyNews	10.55	13.72	8.55	6.17	5.99	5.38
Stella	IMDb	2.42	–	–	1.63	–	–
BGE-M3	IMDb	–	–	–	–	–	–
Non-contrastive / fine-tuned models							
Roberta-Large	IMDb	5.09	5.45	5.20	1.56	4.55	3.58
Roberta-Large	TwentyNews	28.03	27.24	27.00	35.57	34.15	20.53
InBedder	IMDb	4.12	4.93	5.20	2.48	2.97	2.81
InBedder	TwentyNews	39.53	28.34	30.13	34.88	31.96	18.54

Table 15: Mean retention (%) at budget=256 for gradient, activation variance, and learned mask selection vs. random, across 14 representative tasks. Bold indicates best result.

Model	Method	Classification (mean of 2)	Clustering (mean of 2)	Retrieval (mean of 2)	STS (mean of 2)
Contrastive models					
GTE-Large	Random	**99.1**	89.3	91.0	99.3
GTE-Large	Gradient	98.9	94.1	92.3	98.2
GTE-Large	Act. Var.	98.7	**95.6**	**93.8**	**99.7**
GTE-Large	Mask	97.8	88.5	90.5	98.1
Stella	Random	**98.8**	90.5	93.2	98.4
Stella	Gradient	98.5	93.8	**94.6**	97.9
Stella	Act. Var.	97.8	**94.2**	93.1	**98.8**
Stella	Mask	97.2	89.1	92.3	97.6
BGE-M3	Random	96.3	87.2	85.5	**97.8**
BGE-M3	Gradient	96.1	88.6	87.1	97.0
BGE-M3	Act. Var.	95.8	**89.4**	**88.2**	97.5
BGE-M3	Mask	95.4	86.0	84.7	96.2
Non-contrastive models					
Roberta-Large	Random	99.5	**80.5**	**106.3**	**108.5**
Roberta-Large	Gradient	97.8	73.2	85.1	93.4
Roberta-Large	Act. Var.	96.2	72.1	84.5	92.4
Roberta-Large	Mask	**99.8**	78.3	102.1	105.2
InBedder	Random	97.2	80.1	99.5	96.3
InBedder	Gradient	94.8	71.5	88.3	85.2
InBedder	Act. Var.	93.3	68.2	87.6	84.3
InBedder	Mask	**97.6**	**80.8**	**100.8**	**97.1**

Table 16: Per-task importance definition comparison for Roberta-Large ($w = 4$, budget=256). Spearman correlations and normalized entropy for standalone, LOO, marginal, and Shapley importance. Low standalone-to-alternative correlations confirm importance is context-dependent.

	Spearman ρ with Standalone				Normalized Entropy			
Task	LOO	Marginal	Shapley	p (best)	Stand.	LOO	Marginal	Shapley
IMDb	-0.080	0.252	0.176	<0.001	1.000	0.943	0.942	0.707
Banking77	-0.001	0.078	0.156	0.013	0.996	0.949	0.914	0.860
TwentyNews	-0.038	0.137	0.132	0.029	0.999	0.922	0.950	0.933
Medrxiv	0.018	0.039	0.073	0.243	0.999	0.952	0.945	0.842
STSB	0.387	0.705	0.412	<0.001	0.998	0.343	0.677	0.655
BIOSSES	0.260	0.568	0.459	<0.001	0.985	0.921	0.920	0.783

Table 17: Non-contiguous vs. contiguous selection at budget=64 (win_size=4). Greedy forward allows non-contiguous subsets. Jaccard shows overlap between oracle and greedy selections.

Model	Task	Oracle	Greedy Fwd.	Random	Jaccard
GTE-Large	IMDb	99.9%	102.4%	98.5%	0.10
GTE-Large	Banking77	99.8%	101.2%	97.9%	0.10
GTE-Large	TwentyNews	84.1%	86.2%	67.7%	0.14
GTE-Large	BIOSSES	102.9%	108.6%	92.2%	0.14
Stella	IMDb	100.1%	102.8%	99.3%	0.11
Stella	Banking77	100.2%	101.7%	98.8%	0.12
Stella	TwentyNews	83.7%	87.2%	68.5%	0.15
Stella	BIOSSES	101.5%	107.3%	93.1%	0.13
Roberta-Large	IMDb	103.2%	109.3%	94.3%	0.19
Roberta-Large	Banking77	93.3%	102.3%	87.2%	0.07
Roberta-Large	TwentyNews	93.0%	102.9%	70.4%	0.10
Roberta-Large	BIOSSES	115.9%	146.4%	90.6%	0.19
InBedder	IMDb	101.5%	105.8%	96.3%	0.16
InBedder	Banking77	95.8%	100.3%	89.5%	0.09
InBedder	TwentyNews	87.3%	95.1%	73.1%	0.12
InBedder	BIOSSES	108.6%	128.7%	88.2%	0.18

Table 18: OOD transfer gap (source-optimized retention − random retention) at budget=256. Positive values mean source-optimized helps; negative means random is better. For contrastive models, OOD selection is comparable to random.

Pair	GTE	Stella	InBedder	BGE-M3
NFCorpus → ArguAna	+0.4	−0.8	−7.9	−0.6
NFCorpus → SciFact	+0.6	+1.8	−3.4	−2.7
IMDb → Banking77	−0.4	+1.1	+0.8	−0.9
STSB → BIOSSES	+0.3	−0.0	+8.5	−1.9

Table 19: Random selection distribution statistics across 100 seeds (mean over 20 tasks per model). CV = coefficient of variation. P5 = 5th percentile. CVaR-5 = conditional value at risk at 5%.

Model	Budget	Mean ± Std	CV	P5	CVaR-5
Contrastive models					
GTE-Large	256	97.5 ± 0.6	0.6%	96.6	96.5
GTE-Large	64	92.9 ± 1.4	1.5%	90.8	90.6
GTE-Large	16	58.7 ± 3.5	6.0%	53.1	52.6
Stella	256	97.1 ± 0.5	0.5%	96.3	96.2
Stella	64	93.9 ± 1.1	1.2%	92.1	91.9
Stella	16	62.4 ± 2.8	4.5%	57.9	57.4
Non-contrastive models					
Roberta-Large	256	102.9 ± 0.8	0.8%	101.6	101.4
Roberta-Large	64	92.3 ± 2.1	2.3%	88.7	88.2
Roberta-Large	16	43.5 ± 4.2	9.7%	36.9	35.8
BART-Base	256	97.3 ± 0.9	0.9%	95.8	95.6
BART-Base	64	85.6 ± 2.0	2.3%	82.3	81.8
BART-Base	16	40.1 ± 3.8	9.5%	34.2	33.4